

SYEDA ATHER FATIMA

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Johar Town, Lahore

OBJECTIVE

Motivated Information Technology graduate with a strong interest in Machine Learning and Python development, seeking an internship or entry-level role where I can apply hands-on project experience in AI/ML, API integration, and computer vision. Quick learner, team-oriented, and eager to contribute in collaborative, innovation-driven environments.

EDUCATION

University of Education

Bachelor of Science in Information Technology

Lahore, Pakistan

2022 – 2026

- CGPA: **3.81 / 4.00**

- Relevant Coursework: Machine Learning, Computer Vision, Data Structures, Database Systems, Software Engineering

Punjab Group of Colleges

FSc Pre-Engineering

Lahore, Pakistan

2020 – 2022

- Score: **1010 / 1100**

WORK EXPERIENCE

Arch Technologies

Machine Learning Engineer Intern

Remote

1st March 2026 – 30 April 2026

- Contributed to ML model development and deployment pipelines in a remote collaborative environment.
- Assisted in data preprocessing, feature engineering, and model evaluation workflows.
- Collaborated with senior engineers on integrating ML components into production-ready systems.

OASIS INFOBYTE

Python Developer Intern

Remote

November 2025- Jan 2026

- Completed a one-month virtual internship focused on building Python-based applications.
- Developed a **Random Password Generator**, **BMI Calculator**, and a **Weather App** using REST API integration (OpenWeatherMap).
- Managed task submission via GitHub with professional documentation and LinkedIn-based progress reporting.

PROJECTS

Diabetic Retinopathy Segmentation & Severity Classification with XAI | UNet++, PyTorch, GradCAM

2025

- Built a two-stage pipeline using **UNet++** for retinal lesion segmentation (microaneurysms, hemorrhages, exudates) followed by a severity classifier for grading No DR through Proliferative DR.
- UNet++'s dense skip connections improved boundary delineation of fine retinal structures compared to vanilla UNet.
- Integrated Explainable AI (XAI) via GradCAM and SHAP to generate clinically interpretable heatmaps; evaluated explanation quality using causal consistency and faithfulness tests.

Skin Disease Analysis with DermNet | EfficientFormer, ConvNeXt, DINOv2, Vision Mamba

2025

- Conducted a systematic benchmark of five architectures for multi-class skin condition classification on the DermNet dataset: **EfficientFormer-L1 (12M)** and **L3 (26M)**, **ConvNeXt-Base (198M)**, **DINOv2 ViT-B/14 (86M)**, and **Vision Mamba (89M)**.
- Compared model efficiency vs. accuracy trade-offs across lightweight mobile-friendly and large-scale vision transformer architectures.
- Applied data augmentation and class-balancing strategies to handle severe class imbalance; evaluated with per-class F1, macro accuracy, and confusion matrices.

Diet Recommendation System | Python, ML, Pandas, scikit-learn

2024

- Built a personalized dietary recommendation engine incorporating BMI, health goals, allergies, and nutritional constraints.
- Implemented content-based filtering with nutritional clustering to generate balanced meal plans.
- Designed a clean CLI/web interface for real-time user interaction and plan generation.

Fine-Tuned Qwen 2.5 1.5B on English Quotes Dataset | QLoRA, LoRA, HuggingFace

2024

- Fine-tuned the Qwen 2.5 1.5B language model on the English Quotes dataset from HuggingFace using QLoRA and LoRA for parameter-efficient training.

- Reduced GPU memory footprint by 60%+ vs full fine-tuning while maintaining generation quality.
- Evaluated output coherence and style consistency across motivational, literary, and philosophical quote categories.

Python Mini Applications (OIBSIP) | Python, REST API 2024

- Built a weather app consuming the OpenWeatherMap API with real-time city-level forecasts; random password generator with configurable entropy; and a BMI calculator with health-range classification.
- Task 1 Task 2 Task 3

TECHNICAL SKILLS

Languages: Python, HTML, CSS, SQL

ML / AI: PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, LoRA/QLoRA, Transfer Learning

Computer Vision: OpenCV, CNNs, Vision Transformers (ViT), YOLO, GradCAM, SHAP

NLP / LLMs: Fine-tuning LLMs, Prompt Engineering, Text Classification, Qwen 2.5

Data & Tools: Pandas, NumPy, Matplotlib, Git, GitHub, Jupyter Notebook

Productivity: Microsoft Word, PowerPoint, Excel

Languages (Human): English (Proficient), Urdu (Native), Punjabi (Native)

ACHIEVEMENTS & AWARDS

Honehar Scholarship Punjab, Pakistan
Chief Minister of Punjab 2022

- Awarded the Government of Punjab's merit-based scholarship for outstanding academic performance in FSc Pre-Engineering.

PM Laptop Punjab, Pakistan
Prime Minister of Pakistan 2022

- Awarded the Government of Pakistan's merit-based PM laptop for outstanding academic performance in my Bachelor's degree.